



SCHOOL OF MATHEMATICS
AND PHYSICS

Physics in Performance

12 October 2018

Research Plan for the 3rd Year Module
PHY3003M Physics Project

My project, Physics in Performance, ties together both physics and drama to create a useful resource and way of conveying physics. During the first semester, I will research a topic and during the second semester, I will create a video which explains some of the principles from my research during first semester. This may include an experiment to show the key elements and participation from the surrounding community.

After researching previous people who have worked within these two fields, I have found a couple of people who have produced work similar to the way I intend my final piece to look. I found a YouTube channel called 'Veritasium' which contains an extensive series of videos produced by Derek Muller about physics of the world. I have also found a lady, namely Dr Jessamyn Fairfield who incorporates comedy and physics in one in order to engage her audience as well as teaching different disciplines of physics. This interests me as stand-up comedy is a passion of mine and finding a way to incorporate physics into a comedy sketch would be a potential idea for my performance. I intend to get in contact with both these individuals to determine whether they have any ideas to help with my project.

I have decided to do this project with the aim of using it to further my career as an actor. I feel that by combining both performance and physics for my final year project will open doors to either discipline once I have finished.

To continue with my project, I will require a topic in which to create the video about. For this I will require a set of books related to physics containing subjects that will be able to be turned into a performance. I have found three books so far that I feel will offer plenty of options for my topic. These include: The Physics of Everyday Things by James Kakalois, The Physics of Sound by David Stork and Richard Berg, and Classical Mechanics by T.W.B. Kibble. From these books I will aim to find the topic in which I will research and then, during second semester, create my performance piece around. However, I may require further literature to find my topic and once I've found my topic, to help with research into the chosen topic.

For references I will be using:

- <https://plus.maths.org/content/career-interview-actor-and-mathematician>
- <https://lommerde-tonen.com/project/higgs-stand-up-physics/>
- <https://www.irishexaminer.com/ireland/stand-up-scientist-uses-comedy-to-popularise-physics-454912.html>
- The Physics of Everyday Things, James Kakalois.
- The Physics of Sound, David Stork and Richard Berg.
- Classical Mechanics, T.W.B. Kibble.

Once my topic has been chosen I will search for more literature to enhance my knowledge about the topic to help with my understanding.

The work place in which I will be doing most of my project will be in the University of Lincoln, Isaac Newton building. Once my topic has been chosen I will have a clearer idea as to where will be beneficial to be situated in order to complete my project.

For equipment and facilities, I will require to complete my project, I will have to determine the topic prior to this as this will inform me of a potential experiment and therefore what equipment I require.

The cost of my project is unclear as of yet as I haven't got a list of required items. Once my topic is decided and a list has been compiled of potential items that I require, this will be discussed in a meeting with my supervisor, Fabien, and an agreement of the overall cost will be arranged.

Action Plan:

Target Area	Problem	Improvement Method	Support required
Deciding the topic of my project.	I cannot continue with my project until its decided.	Read through chosen literature and find a topic which will work well with the performance element of my project.	N/a.
Deciding what items I require for my project.	If I choose to do a practical that I haven't done before I won't know what items I need.	I will speak to my supervisor to decide what I will require.	My supervisor.

Risk Assessment:

RISK ASSESSMENT SUMMARY SHEET (School of Mathematics and Physics July 2017)



1. Location

University of Lincoln Campus: Brayford Pool	Assessment Date: 12/10/18
School/College: Science	Re-assessment Date: 12/01/19
Department/Faculty: Mathematics and Physics	
Activity: Physics Project	
Building/Area: Isaac Newton Building	
Assessors Name: [REDACTED]	
Accountable Manager Name: [REDACTED]	
Health and Safety Liaison Officer (HSLO) Name: [REDACTED]	

2. Details of further action necessary to control risk (with dates)

Task	Action	Who is Responsible	Date

3. Summary of risks (with controls in place)

Assessment of risk	Low	Medium	High	Very High

Risk to Pregnant Workers?	Yes	No	X
Or to Disabled Workers?	Yes	No	X

4. Evaluation

This assessment is an accurate statement of the known hazards, risks and precautions. I certify that the control measures will prevent or, if this is not possible, control the risk subject to the level shown in section 3 (above) and that staff will be adequately trained and supervised, and the identified control measures implemented. The contents of this assessment will be communicated to staff and all relevant persons.

Signature of Assessor: [REDACTED]	Date: 12/10/18
Signature of Accountable Manager (if not Assessor):	Date:
Signature of HSLO (indicating correct procedure followed):	Date:

Probability of Injury/Loss/Harm (P)

1	Very Unlikely
2	Possible
3	Probable
4	Very Likely

Severity of Injury/Loss/Harm (S)

1	Minor	Mild bruising, minor cuts, mild chemical irritation to eyes or skin. No absence from work or absence of less than 3 days.	Minor property damage
2	Serious	Loss of consciousness, burns, breaks or injury resulting in absence from work for more than 3 days. Other non-permanent chemical effects.	Serious property damage confined to the workroom or area
3	Major	Permanent disability or other reportable injury or disease.	Major property damage affecting the building
4	Fatal	Death	Property damage affecting the loss of one or more buildings

[illegible][illegible]

My supervisor and I have begun meeting every Thursday to discuss the latest updates with the project and to ask any questions that have arisen during the week leading up to the meeting.